SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA0307	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	T Deekshith	**Reg. No.:**	192472253
Section / Batch:	2024	**Date:**	01-08-2026

Code of Conduct

I T Deekshith(192472253) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: T Deekshith

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

Q1

Scenario Input Words: analyzing, analysis, analytical

Approach:

1. Tokenize the input words.
2. Identify prefixes, roots and suffixes.
3. Apply rule-based morphological decomposition.
4. Classify transformations as inflectional or derivational.
5. Normalize all forms to a common root for indexing and retrieval.

Word	Morphological Breakdown	Rule	Classification	Normalized
analyzing	analyze + ing	remove -ing	Inflectional	analyze
analysis	analyze + sis (related noun)	noun derivation	Derivational	analyze
analytical	analyze + tic + al	remove derivational suffixes	Derivational	analyze

Explanation: The processing module extracts the common lexical root 'analyze'. Inflectional endings change grammatical form without changing meaning, whereas derivational affixes create new words or word classes. After normalization, every variant is indexed using the same base form to improve searching, clustering, sentiment analysis, and document retrieval.

Q2

Scenario Input Words: disagree, agreement, agreeable

Approach:

1. Tokenize the input words.
2. Identify prefixes, roots and suffixes.
3. Apply rule-based morphological decomposition.
4. Classify transformations as inflectional or derivational.
5. Normalize all forms to a common root for indexing and retrieval.

Word	Morphological Breakdown	Rule	Classification	Normalized
disagree	dis- + agree	prefix removal	Derivational	agree
agreement	agree + ment	remove -ment	Derivational	agree
agreeable	agree + able	remove -able	Derivational	agree

Explanation: The processing module extracts the common lexical root 'agree'. Inflectional endings change grammatical form without changing meaning, whereas derivational affixes create new words or word classes. After normalization, every variant is indexed using the same base form to improve searching, clustering, sentiment analysis, and document retrieval.

Q3

Scenario Input Words: govern, government, governance

Approach:

1. Tokenize the input words.
2. Identify prefixes, roots and suffixes.
3. Apply rule-based morphological decomposition.
4. Classify transformations as inflectional or derivational.
5. Normalize all forms to a common root for indexing and retrieval.

Word	Morphological Breakdown	Rule	Classification	Normalized
govern	govern	base	Root	govern
government	govern + ment	remove -ment	Derivational	govern
governance	govern + ance	remove -ance	Derivational	govern

Explanation: The processing module extracts the common lexical root 'govern'. Inflectional endings change grammatical form without changing meaning, whereas derivational affixes create new words or word classes. After normalization, every variant is indexed using the same base form to improve searching, clustering, sentiment analysis, and document retrieval.

Q4

Scenario Input Words: activate, activation, reactivation

Approach:

1. Tokenize the input words.
2. Identify prefixes, roots and suffixes.
3. Apply rule-based morphological decomposition.
4. Classify transformations as inflectional or derivational.
5. Normalize all forms to a common root for indexing and retrieval.

Word	Morphological Breakdown	Rule	Classification	Normalized
activate	activate	base	Root	activate
activation	activate + ion	remove -ion	Derivational	activate
reactivation	re + activate + ion	remove re-, -ion	Derivational	activate

Explanation: The processing module extracts the common lexical root 'activate'. Inflectional endings change grammatical form without changing meaning, whereas derivational affixes create new words or word classes. After normalization, every variant is indexed using the same base form to improve searching, clustering, sentiment analysis, and document retrieval.

Q5

Scenario Input Words: create, creates, creating

Approach:

1. Tokenize the input words.
2. Identify prefixes, roots and suffixes.
3. Apply rule-based morphological decomposition.
4. Classify transformations as inflectional or derivational.
5. Normalize all forms to a common root for indexing and retrieval.

Word	Morphological Breakdown	Rule	Classification	Normalized
create	create	base	Root	create
creates	create + s	remove -s	Inflectional	create

creating	create + ing	remove -ing	Inflectional	create

Explanation: The processing module extracts the common lexical root 'create'. Inflectional endings change grammatical form without changing meaning, whereas derivational affixes create new words or word classes. After normalization, every variant is indexed using the same base form to improve searching, clustering, sentiment analysis, and document retrieval.

Conclusion

Rule-based morphological analysis improves NLP tasks by decomposing words into prefixes, roots, and suffixes. Normalizing related word forms reduces redundancy, improves retrieval accuracy, and ensures semantic consistency across indexing, clustering, and text analysis applications.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							

4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________